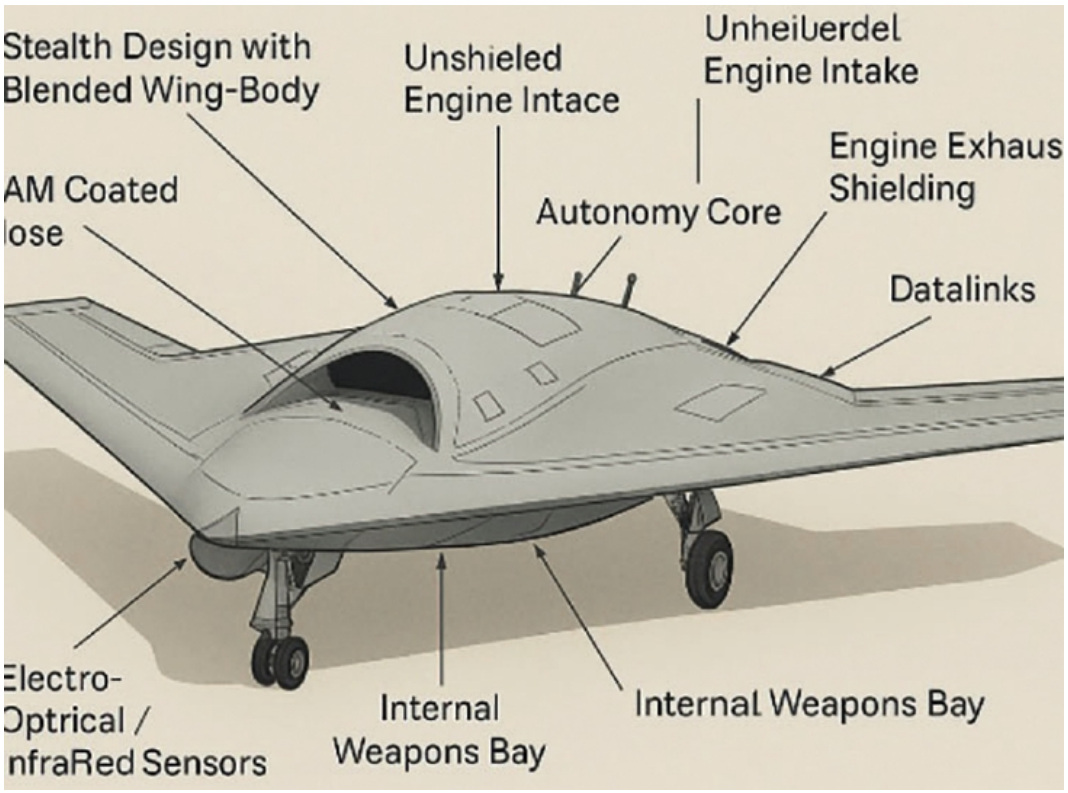


Figure 5.10 Okhotnik-B (Hunter-B) (2020).⏎

5.7 Challenges and Limitations of Fourth-generation Drones

While fourth-generation drones represent a monumental leap in autonomy, stealth, and networked warfare, their development and operational deployment present a range of technological, strategic, ethical, and logistical challenges. These limitations stem not only from the complexity of integrating AI and machine learning (ML) with aerospace systems but also from issues surrounding cost, reliability, cybersecurity, and global regulations. The following section outlines the primary challenges impeding the full-scale deployment and optimization of fourth-generation UAV systems.

5.7.1 Technological challenges

The most significant hurdle lies in achieving true cognitive autonomy, where drones can interpret high-level mission objectives and adapt to